Quadratic contractions of the memory core
An exact adjacent-charge covariance formula

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Abstract

We evaluate the ordinary second Gaussian-chaos part of the quartic operator entering the second centered memory coefficient. The answer uses neutral and adjacent-charge embedded-ensemble covariance maps, retaining the same couplings on both charge sectors. A finite example has a nonzero dynamical residual after removing sample-dependent scalar parts and the previously available quadratic descendants. The formula supplies the missing finite-N contraction interface; its double-scaled residual strength is the next calculation.

1 Model, target and the correct Gaussian subtraction

Use standard CAR and the full overlapping complex Gaussian p-body Hamiltonian. The couplings obey $\mathbb{E}J_{IJ}J_{KL} = w\delta_{IL}\delta_{JK}$. All results here are exact at finite N, fixed charge $m \geq p$, with annealed normalized Hilbert–Schmidt inner product $\langle X, Y\rangle_m = \mathbb{E}\,\mathrm{tr}_m(X^\dagger Y)$. Take $\hbar = 1$. The physical variance is $w = 4^p(J_s^2 N/p^2)(p!/N^p)^2$; explicit finite tests use $w = 1$. No large-N or small-λ approximation is used below.

Write $\mathcal{L} = [H, \cdot]$ and

$$A = \sum_i (\mathcal{L}c_i)^\dagger \mathcal{L}c_i, \qquad \mathcal{E} = \sum_i (\mathcal{L}^2 c_i)^\dagger \mathcal{L}^2 c_i + \tfrac{1}{4}\mathcal{L}^2 A. \tag{1}$$

The preceding memory calculation identified $\mathcal{E}$ as the core of mZ_2 after its explicit first-return counterterm, up to a vanishing double-scaled remainder. This note evaluates its lower coupling chaos.

Choose independent real Gaussian coordinates x_a with variances v_a, so that $H = \sum_a x_a V_a$. For any homogeneous quartic matrix polynomial P, put $\Delta = \sum_a v_a \partial_{x_a}^2$ and

$$Q_P = \tfrac{1}{2}\Delta P, \qquad e_P = \mathbb{E}P. \tag{2}$$

The Gaussian normal-ordering identity gives exactly

$$W_0 P = e_P, \qquad W_2 P = Q_P - 2e_P, \qquad W_4 P = P - Q_P + e_P. \tag{3}$$

Indeed $\Delta^2 P = 8e_P$ and $\mathbb{E}Q_P = 2e_P$. These are coordinate-chaos projections. The q-Hermite Wick polynomials of the spectral law in [2] have a different definition; an equality between them requires a separate limit argument.

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2 Charge removal and the covariance primitives

On neighboring sectors write $h = H_m$, $k = H_{m-1}$ and $c_i : \mathcal{H}_m \to \mathcal{H}_{m-1}$. Define

$$\mathcal{R}(X) = \sum_i c_i^\dagger X c_i, \quad R_j = \mathcal{R}(k^j), \quad R_0 = mI, \quad R_1 = (m-p)h. \tag{4}$$

Directly expanding the two fermion derivatives gives

$$A = R_2 + (2p-m)h^2, \tag{5}$$
$$\mathcal{E} = R_4 - 2\{h, R_3\} + \tfrac{5}{4}\{h^2, R_2\} + \tfrac{7}{2} h R_2 h + (4p-3m)h^4. \tag{6}$$

For example the square of $k^2 c_i - 2kc_i h + c_i h^2$ yields the coefficients $1, -2, 1, 4, 4p-3m$ before adding $\mathcal{L}^2 A/4$.

Let (u, v) be an independent copy of (h, k), using the same independent couplings on the two sectors. Set

$$\Phi_+(X) = \mathbb{E}_u[uXu], \qquad \Phi_-(X) = \mathbb{E}_v[vXv], \tag{7}$$
$$\Psi(X) = \mathbb{E}_{u,v}[uXv], \qquad \Gamma(X,Y) = \sum_i \Psi(c_i^\dagger X) Y c_i. \tag{8}$$

Here Ψ acts on $d_m \times d_{m-1}$ matrices, while X, Y in Γ act on $\mathcal{H}_{m-1}$. The neutral constants are

$$C = w\binom{m}{p}\binom{N-m+p}{p}, \quad C_- = w\binom{m-1}{p}\binom{N-m+p+1}{p}, \quad a = (2p-m)C + mC_-. \tag{9}$$

Thus $\Phi_+(I) = CI$, $\Phi_-(I) = C_- I$, $\mathbb{E}A = aI$. All binomial coefficients outside their range vanish.

These are the standard embedded-ensemble covariance operations [1, 2], not new independent random probes. If $T_{IJ}^{(r)} = (a_I^\dagger a_J)|_{\mathcal{H}_r}$ denotes the charge-r block, the mixed map is

$$\Psi(X) = w \sum_{I,J} T_{IJ}^{(m)} \, X \, T_{JI}^{(m-1)}. \tag{10}$$

Neutral maps are obtained by using the same block on both sides. In a neutral irreducible rank-ν channel,

$$\Phi_m(O_\nu) = \kappa_\nu(m) O_\nu, \qquad \kappa_\nu(m) = w\binom{m-\nu}{p}\binom{N-m+p-\nu}{p}. \tag{11}$$

The cross-sector map in (10) must be retained before taking an asymptotic limit.

3 The closed single-contraction formula

For a polynomial $P(h,k)$ let $\mathcal{C}P = \mathbb{E}_{u,v}[t^2]P(h+tu, k+tv)$. This equals Q_P in (2). Introduce

$$P_+ = \Phi_+(h), \qquad P_- = \Phi_-(k), \tag{12}$$

$$U = \Gamma(k, I) + \Gamma(I, k), \tag{13}$$

$$V = \Gamma(I, k^2) + \Gamma(k, k) + \Gamma(k^2, I), \tag{14}$$

$$U_h = \sum_i [\Psi(hc_i^\dagger k)c_i + \Psi(hc_i^\dagger)kc_i]. \tag{15}$$

Every quantity on the right contains only the original h, k and the deterministic covariance maps. The five contractions in (6) are

$$D_4 := \mathcal{C}R_4 = \mathcal{R}(3C_-k^2 + \{k, P_-\} + \Phi_-(k^2)), \tag{16}$$

$$D_{13} := \mathcal{C}\{h, R_3\} = \{h, \mathcal{R}(2C_-k + P_-)\} + V + V^\dagger, \tag{17}$$

$$D_{22} := \mathcal{C}\{h^2, R_2\} = 2CR_2 + 2mC_-h^2 + hU + U_h + U^\dagger h + U_h^\dagger, \tag{18}$$

$$D_{121} := \mathcal{C}(hR_2h) = \Phi_+(R_2) + Uh + hU^\dagger + mC_-h^2, \tag{19}$$

$$D_{40} := \mathcal{C}h^4 = 3Ch^2 + \{h, P_+\} + \Phi_+(h^2). \tag{20}$$

Consequently the answer is the Hermitian quadratic polynomial

$$\boxed{Q_\mathcal{E} = D_4 - 2D_{13} + \tfrac{5}{4}D_{22} + \tfrac{7}{2}D_{121} + (4p - 3m)D_{40}.} \tag{21}$$

The scalar $e = \mathbb{E}\mathcal{E} = e_m I$ follows either by a second contraction or by the original fourth moment. Equation (3) then recovers $W_2\mathcal{E}$ and $W_4\mathcal{E}$ exactly.

To check the ordering, write $R_2(h+tu, k+tv) = R_2 + tB + t^2\mathcal{R}(v^2)$ with $B = \mathcal{R}(kv + vk)$. Then $\mathbb{E}uB = U$, $\mathbb{E}uhB = U_h$, and $\mathbb{E}Bu = U^\dagger$. These three expectations generate the mixed terms in D_{22} and D_{121}. For R_3, the coefficient linear in v is $\mathcal{R}(vk^2 + kvk + k^2v)$; its left contraction with u is V. This proves the mixed formulas without an independence assumption between neighboring sectors.

The formula is also an interface for restoring the low-chaos parts of the four-slot descendant basis: all of its members are linear combinations of the five raw words in (6).

4 Projection onto old directions and a dynamical residual

Put $B = h^2 - CI$ and $O = A - aI$. The exact projection coefficient onto the energy quadratic is

$$c_E = \frac{\langle B, W_2\mathcal{E}\rangle_m}{\langle B, B\rangle_m}. \tag{22}$$

Also retain $J_hF = \{h, F\}/2$, where $F = W_1T_3 - \bar{t}h$ and $\bar{t} = \langle h, W_1T_3\rangle_m/C$ are the previously defined linear rank-filtered field and its energy projection. For direct evaluation in the same covariance primitives,

$$U_A = U + (2p - m)(P_+ + Ch), \tag{23}$$

$$W_1T_3 = \mathcal{R}(2C_-k + P_-) + (3p - m)(2Ch + P_+) - 3(ah + (U_A + U_A^\dagger)/2). \tag{24}$$

All Gram entries can be evaluated without disorder sampling. If a Hermitian quadratic polynomial has symmetric coefficients $P = \sum_{ab} x_a x_b P_{ab}$, then

$$\langle W_2P, W_2S\rangle_m = 2\sum_{ab} v_a v_b \,\mathrm{tr}_m(P_{ab}^\dagger S_{ab}). \tag{25}$$

At finite N, an additional conserved piece can be a random scalar matrix. Remove it sample by sample with $\pi X = X - \mathrm{tr}_m(X)I$. Project $\pi W_2\mathcal{E}$ onto the span of $\pi B, \pi O, \pi J_hF$ and call the residual $\mathcal{D}$. This removes a specified conserved subspace, not the full commutant. Exact results at $w = 1$ are

(N, p, m)	e_m	c_E	$\langle\mathcal{D}, \mathcal{D}\rangle_m$
$(3, 1, 2)$	38	5	0
$(4, 2, 3)$	684	1089/17	0
$(5, 2, 3)$	2268	4596/41	248945355/14903

The coefficient c_E refers to the unmodified B projection; the last column includes π and all three old directions. For the last system their coefficients are, respectively,

$$\left(\frac{2106991}{44709}, \frac{1499753}{29806}, -\frac{77111}{89418}\right). \tag{26}$$

The saved deterministic integer coupling vector gives

$$\mathrm{tr}_m[h, \mathcal{D}]^\dagger[h, \mathcal{D}] = \frac{12991771132439678939}{2220994090} > 0. \tag{27}$$

Thus the extra finite-N residual is genuinely dynamical in this example. A positive finite residual alone does not determine its double-scaled norm or spectral measure.

5 Analytic control and the next asymptotic calculation

There is an all-N one-body control. For $p = 1$, let J be the fundamental $N \times N$ GUE matrix and $H = d\Gamma(J)$. The CAR give $\mathcal{L}^2 c = J^2 c$, so

$$A = d\Gamma(J^2), \quad [H, A] = 0, \quad \mathcal{E} = d\Gamma(J^4). \tag{28}$$

At $w = 1$, $\mathbb{E}KXXK = \operatorname{tr}(X)I$ for the fundamental independent matrix K, where this trace is unnormalized. The six contractions give

$$Q_{\mathcal{E}} = 3NA + 2\operatorname{tr}(J)H + m\operatorname{tr}(J^2)I, \qquad e_m = m(2N^2 + 1). \tag{29}$$

Every term commutes with H. This proves why a small one-body example cannot establish a new dynamical mode, even when an incomplete Gram subtraction leaves a positive norm.

The saved verification comprises three full finite-sector Gram calculations (45, 666 and 5050 symmetric coefficient blocks), 807 independent quartic polarization directions, and 56 one-body identities in 14 charge sectors. Polarization evaluates the original commutator-defined core, independently of the charged covariance expression. Gaussian coordinate variances are 1 on diagonal couplings and 1/2 on each real or imaginary off-diagonal coordinate. The Gram sums use rational arithmetic after checking dyadic matrix coefficients; no Monte Carlo fit or charge truncation is used.

Restoring the physical variance rescales $W_2\mathcal{E}$ by w^2, its squared norm by w^4, and the commutator witness by w^5. For the intended double-scaled problem, first set $p^2/N \to \lambda > 0$ at the physical w, then study small λ. The exact maps in (21) should now be decomposed in their irreducible input and output ranks. In particular, the $O(p)$ and $O(m)$ coefficients must be combined before discarding covariance differences.

The immediate target is the limiting norm and mixed energy moments of the residual after restoring all fourth-chaos descendants. That will decide whether it vanishes, joins existing filtered quadratic channels, or supplies another direction. Next come its full two-energy measure, the genuine Z_2 spectrum, and the covariance at two finite memory parameters. Spatial reconstruction and the bulk mass dictionary remain the subsequent physical objectives. No finite-N channel count is being identified here with a de Sitter particle spectrum.

References

[1] V. K. B. Kota, *Bivariate moments of the two-point correlation function for embedded Gaussian unitary ensemble with k-body interactions*, arXiv:2208.11312v2, Sec. IV, Eqs. (29)–(32).

[2] J. Tall and S. Tomsovic, *Double-scaled bosonic and fermionic embedded ensembles, complex SYK, and the dual Hilbert space*, arXiv:2604.14522v2, Secs. 2.1 and 3.1.